

Mohammad Saifullah Khan

+91-7225867555 | mohammad21@iiserb.ac.in | [mskhan0077.github.io](https://github.com/mskhan0077)

 [LinkedIn](#) |  [mohammad21saif](#) |  [mskhan0077](#) |  [Google Scholar](#)

Bhopal, Madhya Pradesh - 462036, India

RESEARCH INTERESTS

Field & Marine Robotics · Multi-Robot Systems · Informative Path Planning · Decision-making under uncertainty

EDUCATION

- **Indian Institute of Science Education and Research Bhopal** Dec 2021 - June 2026
Integrated BS-MS (Master's level degree) Electrical Engineering and Computer Science with minor in Data Science Bhopal, India
 - CGPA: 8.68/10.00
- **Holy Family Convent Sr. Sec. School** May 2021
Secondary Education Bhopal, India
 - CGPA: 94.2/100.0

SKILLS

- **Programming:** Python, C/C++, MATLAB
- **Robotics & Middleware:** ROS/ROS2, Nav2
- **Embedded Systems & Hardware Integration:** Pixhawk (PX4/ArduPilot), Raspberry Pi, NVIDIA Jetson, Arduino
- **Perception & Sensors:** Cameras, LiDAR, GPS, IMU, Sonar/Echosounder
- **Planning & Control:** Search-based planning, sampling-based planning, kinematic control, information-driven exploration
- **Simulation:** Gazebo, Unity
- **Design & Fabrication:** CAD, 3D printing
- **Machine Learning:** Deep Reinforcement Learning, Probabilistic Modelling, Gaussian Processes
- **Other Tools & Technologies:** Git, Docker, Linux, TCP/IP Networking
- **Languages:** Hindi (native), English (fluent)

RESEARCH EXPERIENCE

- **Multi-Robot Autonomy (MOON) Lab, IISER Bhopal** July 2026 – present
Project Associate I Bhopal, India
 - Developing a bio-inspired surface vehicle for autonomous navigation in vegetation-choked and debris-laden waterways, where conventional propellers foul.
 - Building a Unity-based simulation environment for the lab's marine, ground and aerial platforms.
 - Investigating sim-to-real transfer of learned navigation policies onto the lab's ASV fleet, addressing problems such as hydrodynamic mismatch, sensor noise, actuation delay, etc.
- **Multi-Robot Autonomy (MOON) Lab, IISER Bhopal** June 2025 – June 2026
Graduate Researcher Bhopal, India
 - Developed MA-BaTS, a multi-agent adaptive bathymetric mapping framework using Gaussian Process field reconstruction with Thompson Sampling acquisition; achieved ~68% reduction in traversal distance at equivalent reconstruction error, validated on a two-USV team in outdoor field trials.
 - Built and ran the full deployment stack for these trials with ROS coordination, echosounder and GPS/IMU integration, and mission execution on water.
- **Department of EEE, IIT Guwahati** Mar 2025 – Oct 2025
Visiting Student Researcher (concurrent with MOON Lab) Guwahati, India
 - Visiting researcher at the Energy Resilience Research Lab under Dr. Prabir Barooah.
 - Built an autonomous field-testing platform and conducted real-world experimental validation.
- **Multi-Robot Autonomy (MOON) Lab, IISER Bhopal** Aug 2023 – Apr 2025
Undergraduate Researcher Bhopal, India
 - Co-developed a search-based collision avoidance strategy encoding COLREGs maritime traffic rules for autonomous surface vessels for enabling rule-compliant safe navigation.
 - Prototyped autonomous robotic systems covering sensing, control, deployment, and field trials.

PROJECTS

MA-BaTS: Multi-agent Adaptive Bathymetry Mapping using Thompson Sampling

June 2025 – Present

MOON Lab, IISER Bhopal

- Developed a learning-based adaptive framework for collaborative bathymetric mapping, combining Gaussian Process reconstruction on the seafloor depth field with multi-robot decision making and control under limited sensing.
- Achieved a $\sim 68\%$ reduction in traversal distance at equivalent reconstruction error; framework validated in real-world field deployments.

EcoBoat: Autonomous Cleaning Boat (v2)

Dec 2024 – Feb 2025

MOON Lab, IISER Bhopal

- Developed an autonomous lake-cleaning robot with Tesla-valve-inspired passive debris collection.
- Implemented random-sampling-based coverage planning for debris collection missions.

AquaRP: Autonomous Surface Vehicle (ASV) Research Platform

Aug 2024 – Sep 2024

MOON Lab, IISER Bhopal & IIT Guwahati

- Designed and integrated a low-cost (\$2200) autonomous surface vehicle with embedded control (Pixhawk), onboard computation (NVIDIA Jetson), and multi-modal sensing (camera, GPS/IMU, echosounder).
- Implemented autonomous navigation and multi-agent coordination, enabling collaborative field missions and real-world system validation.

Manipulator Arm for Pick-and-Place Task

Dec 2023 – Jul 2024

MOON Lab, IISER Bhopal

- Designed a custom 5-DOF manipulator; implemented kinematic control using Robotics Toolbox.
- Integrated vision-based object pose estimation using FoundationPose.

Autonomous Cleaning Boat

May 2023 – Jul 2023

MOON Lab, IISER Bhopal

- Developed an autonomous debris-cleaning robot with YOLOv5s-based detection in VRX simulation.
- Deployed onboard computation using Jetson Nano with Pixhawk-based navigation.

GesTr: Wearable Indian Sign Language Translation Device

Dec 2022 – Mar 2023

ED/ICT Lab, IISER Bhopal

- Developed a wearable ISL-to-text/audio system with 97% real-time accuracy; shortlisted at 2022-23 IEEE CASS SDC.

HONORS AND AWARDS

Bharat Innovates 2026 (National Basecamp)

Dec 2025

IIT Gandhinagar

- Shortlisted from 1,400 teams to the top 400 to showcase the GesTr.

Best Oral Presentation (EECS Symposium)

Feb 2025

Department of EECS, IISER Bhopal

- Presented a marine robotics platform for in-field experimentation and its application in water body cleaning.

Inclusion@RSS 2024 Fellowship

July 2024

Robotics: Science and Systems Foundation

- Awarded fellowship to attend the Robotics: Science and Systems (RSS) 2024 Conference at TU Delft, the Netherlands.
- Selected as one of 27 students globally for this prestigious fellowship.

ISRO Robotics Challenge-URSC 2024 (IRoC-U2024)

July 2024

Indian Space Research Organisation (ISRO)

- Successfully qualified Hardware Functionality Demonstration and Rover Functionality Demonstration rounds.
- Selected as one of the top 20 teams in India.

IASc-INSa-NASI Summer Research Fellowship

May 2024

Indian Academy of Sciences (IASc), INSA & NASI

- Awarded a competitive national summer research fellowship to conduct summer research under Dr Nagahanumaiah at CMTI, Bengaluru.

2022-23 IEEE CASS Student Design Competition

Mar 2023

IEEE CASS

- Achieved third position in Asia-Pacific.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, T=THESIS

- [C.1] K. Jaiswal, **M. S. Khan**, R. Kulkarni, S. P. Baliyarasimhuni, "A Search-Based Collision Avoidance Strategy with COLREGs for Autonomous Surface Vehicles," IEEE International Symposium on Multi-Robot and Multi-Agent Systems (MRS), 2025. **(Oral Presentation)**

PATENTS

- [1] **A User-Wearable Indian Sign Language Translator.** Indian Patent Office, Application number: 202321071011, Date of Filing: 18/10/2023

TEACHING / OUTREACH

- **Department of EECS, IISER Bhopal** Aug 2024 – Nov 2024
Teaching Assistant Bhopal, India
 - Teaching assistant for ECS 203 (Basic Electronics); led tutorials and supported circuit analysis and lab concepts.
- **Ashoka University** Feb 2026
Invited Workshop Speaker Sonapat, India
 - Gave a hands-on introduction to robotics and ROS2 to undergraduate CS and Physics students.

REFERENCES

1. **Dr Sujit Pedda Baliyarasimhuni**
Professor, Department of Electrical Engineering and Computer Science
Indian Institute of Science Education and Research Bhopal
Email: sujit@iiserb.ac.in
Phone: +91 755 269 2649
2. **Dr Pydi Ganga Bahubalindrani**
Associate Professor, Department of Electrical Engineering and Computer Science
Indian Institute of Science Education and Research Bhopal
Email: ganga@iiserb.ac.in
Phone: +91 755 269 2650